Michelle Ding

michelle.ding01@utexas.edu, cs.utexas.edu/~mding01/, linkedin.com/in/michelleding01

Education

2025 - Present **The University of Texas at Austin**
PhD in Quantum Computing
Advised by Scott Aaronson

2021 - 2025 **The University of Texas at Austin**
BS in Computer Science, BS in Mathematics, and Minor in Business
CS Turing Scholars Honors Program

Research

Sep 30, 2026 Ryan Anselm, MD, Dar Gilboa, Sabee Grewal. **Optimal and Verifiable Quantum Advantages in Communication Complexity** arXiv:2610.05585.

Apr 12, 2025 **Undergraduate Honors Thesis: Peaked Boson Sampling: towards efficiently verifiable and NISQ-able quantum advantage** Manuscript. Poster. *Best poster award.*

Jul 15, 2024 George Tsoukalas, Jasper Lee, John Jennings, Jimmy Xin, MD, Michael Jennings, Amitayush Thakur, Swarat Chaudhuri. **PutnamBench: Evaluating Neural Theorem-Provers on the Putnam Mathematical Competition** *NeurIPS 2024. ICML 2024. Best Paper Award.*

Teaching

Fall 2025 Teaching Assistant for **Intro to Quantum Information Science: Honors** at UT Austin.

Fall 2024 Undergraduate Grader for **Introduction to Quantum Information Science** at UT Austin.

Fall 2023 Teaching Assistant for **Quantum Computing Freshman Research Initiative** at UT Austin.

Spring 2023 Teaching Assistant for **Quantum Computing Freshman Research Initiative** at UT Austin.

Fall 2022 Teaching Assistant for **Discrete Mathematics: Honors** at UT Austin.

Scholarships, Grants & Fellowships

Apr 2026 Texas Quantum Institute (TQI) Graduate Research Fellowship

Aug 2025 McKinley Strahan Graduate Fellowship in Computer Science

Aug 2024 Leonard E. Huber Endowed Scholarship - $3,000

Apr 2022 Freshman Research Initiative Summer Fellowship - $2,000

Aug 2021 Ada Mae Janick Marshall Endowed Presidential Scholarship - $3,500

2020 National Office of Mu Alpha Theta Summer Grant - $2,000

Experience

May 2026 - Aug 2026 Quantum Computing PhD Research Intern at **PNNL** at NYC, New York.
Studied stabilizer one-way communication complexity mentored by Samuel Stein and advised by Scott Aaronson. Researched maximal separations between one-way randomized and quantum communication complexity based on n-qubit stabilizer inputs and measurements.

Jun 2024 - Aug 2024 Research and Development Intern for **Trend Micro** at Austin, TX.
Created structured output logging for internal weekly test automation framework. Relevant tools: Python, MySQL, Docker, Jira, Azure Data Studio.

Nov 2023 - Feb 2024 Research Assistant for the **IFML Lab** at Austin, TX.
Built IFML Retrieval Augmented Generation Models mentored by Adam Klivans. At the Institute for Foundations of Machine Learning at UT Austin. Experimented with LLMs, vector databases, and chatbot RAG algorithms to enhance the accuracy and reliability of generative AI models for identifying NeurIPS papers. Manuscript.

Jun 2022- Jul 2022 Research Assistant at the **J.J. Pickle Quantum Optics Laboratory** at Austin, TX.
Conducted Quantum State Tomography Experiments on PBS mentored by Noah Davis. At the J.J. Pickle Quantum Optics Laboratory, Austin. Used class 3B Laser systems to replicate state tomography and Bell's Inequality experiments. Manuscript.

Community Service

2026 - **Conference reviewer**: *QIP 2026*
2025 - **Journal reviewer**: *Quantum 2025*
Mar 2023 - Aug 2024 Officer / Core Problem Writer for **UT Programming Contest (UTPC)** at UT Austin.
Initiate and implemented written problem editorials which are now publicly available after every contest on Codeforces and Discord. Additionally: write problems for biweekly contests and annual UT Open contest held on the Codeforces platform.

Mar 2023 - Aug 2024 Vice President for the **Mathematics Club** at UT Austin.
Obtained for all students access to free MAA weekly journals sponsored by the Math Department, revived Putnam practice sessions, and created the club's official website.

Skills and Certifications

2024 Editor for the upcoming book *Integrals and Sums Fiesta II*.

2020,2021 Two-time AIME Qualifier at John Foster Dulles High School.

2020 US Figure Skating, Silver Level.

Mar 2020 1st Place Winner of the American Protégé International Music Talent Competition.

Last update: October 2026